

Lesson 8: Intermediate Value Theorem

Continuity of a function will prove to be an important characteristic in many theorems for our course. In order for theorems to be properly applied, we need to meet the certain conditions. Namely, the “if” part, called the **hypothesis**, must be satisfied before we can apply the “then” part, or **conclusion** (consequence) of the theorem.

In this lesson, we will explore our first **existence theorem**. Existence theorems allow us to draw conclusions about a function’s behavior on an interval without precisely locating that behavior.

Topic 1.16: Working with the Intermediate Value Theorem (IVT)

Intermediate Value Theorem

The Big Idea:

The **IVT** says that a continuous function on an interval cannot skip values.

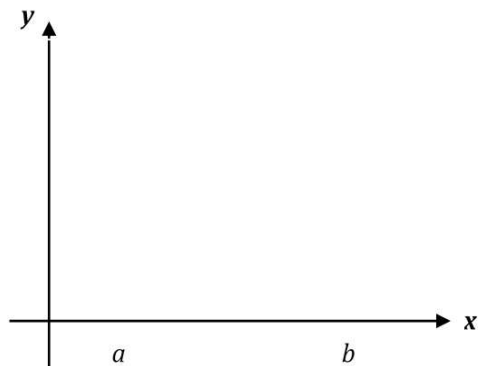
Using If/Then statements:

If a function $y = f(x)$ is continuous on a closed interval $[a, b]$ and $f(a) \neq f(b)$, **Then** $f(x)$ takes on every value between $f(a)$ and $f(b)$ on that interval.

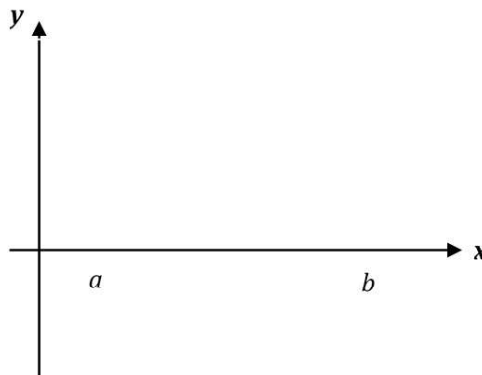
Another Approach:

Said in a different fashion, if we know a y -value, say $y = k$, that resides between the two endpoints, $f(a)$ and $f(b)$, then we are guaranteed at least one x -value, $x = c$, between the endpoints that generates that y -value, such that $f(c) = k$.

EX #1: Graphical representation of the Intermediate Value Theorem and its corollary:

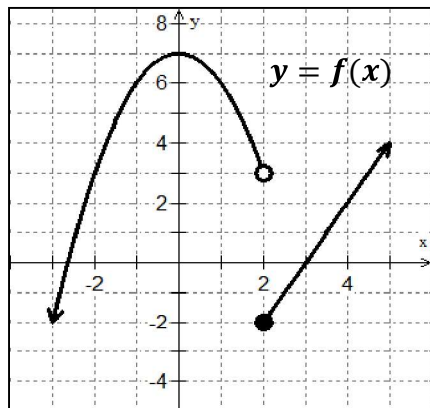


A continuous function on an interval cannot skip values.



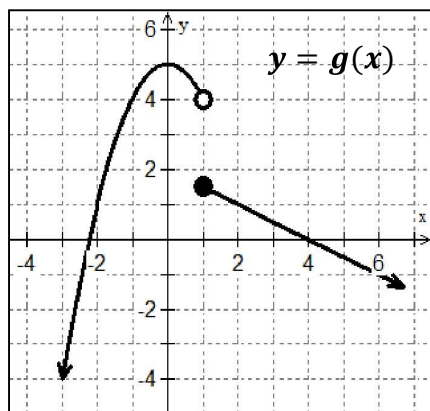
An important outcome of I.V.T. is that it can be helpful in finding zeros of a continuous function on an interval.

EX #2: Does the Intermediate Value Theorem apply on the specified intervals for the piecewise functions shown below? Be sure to explain completely.



A. Is there a value of $x = c$ on the interval $[-2, 5]$ such that $f(c) = 1$?

B. Does I.V.T. guarantee a value of c on the interval $[-2, 5]$ such that $f(c) = 1$? Justify.



C. Is there a value of $x = c$ on the interval $[1, 6]$ such that $g(c) = 3$?

D. Does I.V.T. guarantee a value of c on the interval $[1, 6]$ such that $g(c) = 3$? Justify.

What three conditions are necessary to apply the Intermediate Value Theorem?

1.

2.

3.

EX #3: Apply the IVT, if possible, on $[0, 5]$ so that $f(c) = 1$ for the function $f(x) = x^2 + x - 1$.

EX #4: Use the Intermediate Value Theorem to show that $f(x) = x^3 + 2x - 1$ has a zero in the interval $[0, 1]$.

EX #5: A car travels on a straight track. During the time interval $0 \leq t \leq 60$ seconds, the car's velocity v , measured in feet per second is a continuous function. The table below shows selected values of the function.

t , in seconds	0	15	25	30	35	50	60
$v(t)$ in ft/sec	-20	-30	-20	-14	-10	0	10

A. For $0 < t < 60$, must there be a time t when $v(t) = -5$?

B. Justify your answer.

EX #6: Let $g(x)$ be a continuous function. Selected values of $y = g(x)$ are given in the table below. For which value of k will the equation $g(x) = 3/4$ have **at least two solutions** on the closed interval $[2, 8]$?

x	2	3	4	5	8
$g(x)$	3	4	k	5	2

- (A) 1 (B) $\frac{3}{4}$ (C) $\frac{9}{16}$ (D) $\frac{3}{2}$

EX #7: For the function $f(x) = \begin{cases} (x-3)^2, & x = 5 \\ 6, & 5 < x \leq 10 \end{cases}$. Find $f(5)$ and $f(10)$. Does IVT guarantee a y -value k on $5 \leq x \leq 10$ such that $f(5) < f(k) < f(10)$? Justify your answer.

EX #8: The functions f and g are continuous for all real numbers. The table below gives values of the functions at selected values of x . The function h is given by $h(x) = g(f(x)) + 2$. Explain why there must be a value w for $1 < w < 6$ such that $h(w) = 0$.

x	1	2	6	8
$f(x)$	2	9	8	13
$g(x)$	3	-12	5	28